

Setting instruction of CSU3

CSU3 Setting Manual

Version 0.24

(2025/09/30)

Document History

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2020/03/10	V0.2		1. Chapter-1, link table update 2. Web page image update
2020/04/08	V0.3		1. Chapter-3, content update 2. Web page image update
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2023/04/27	V0.9		1. Chapter-3, add firewall configuration 2. Chapter-3, add maintain server configuration
2023/06/21	V0.10		1. Chapter-3 add export setting.
2023/09/22	V0.11		1. Chapter-3, add account setting. 2. Chapter-3, add Offline Max Charge Current / Power setting.
2023/01/02	V0.12		1. Chapter-2, default login account & password change. 2. Chapter-3, account group description. 3. Appendix add modbus information
2024/06/27	V0.13		1. Appendix add modbus register for input voltage.
2024/07/29	V0.14		1. Offline policy add no charging(End immediately) item. 2. WIFI client add EAP-TLS related configuration item. 3. Backend common section add support ipv6 item 4. Local loading balance move to independent section
2024/11/21	V0.15		1. Add connector operative / inoperative request 2. Add Connector plugged count clear request 3. Add connector plugged count info
2024/12/05	V0.16		1. Chapter 3.1.1 add LCD language configuration 2. Chapter 3.1.2 add Contract install/update·VDV261·R2R·Multi connector charging mode
2025/01/14	V0.17		1. Chapter 4.1.3 update for status code description
2025/02/10	V0.18		1. Chapter 4.1.3 add connector temperatire and present EVCCID information.

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1. Link Personal Computer to EV Charger

Step-1 Power on the EV Charger, and please use a network cable to plug into the EV Charger's CAT 5/5e port, which is back of the EV Charger.

Step-2 Link the network cable to personal computer .

Step-3 Set Ethernet of the personal computer to use static IP, IP address setting value as follow below setting table:

Charger LAN port	PC LAN port	
IP address	IP address	Netmask
192.168.1.10	192.168.1.1	255.255.255.0

Form 1-1 PC static IP address set mapping table

網際網路通訊協定第 4 版 (TCP/IPv4) - 內容

一般

如果您的網路支援這項功能，您可以取得自動指派 IP 設定。否則，您必須詢問網路系統管理員正確的 IP 設定。

☐ 自動取得 IP 位址(O)

☒ 使用下列的 IP 位址(S):

IP 位址(I): 192 . 168 . 1 . 1

子網路遮罩(U): 255 . 255 . 255 . 0

預設閘道(D): . . .

☐ 自動取得 DNS 伺服器位址(B)

☒ 使用下列的 DNS 伺服器位址(E):

慣用 DNS 伺服器(P): . . .

其他 DNS 伺服器(A): . . .

☐ 結束時確認設定(L)

進階(V)...

確定 取消

Image 1-1 PC static IP address setting

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2. Access to setting tool

Access charger configuration website, https://CHARGER_IP_ADDRESS/

- CHARGER_IP_ADDRESS : Please find Table 1-1
- Account : zerova
- Default password : t6sjj4j83

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3. Setting instructions

CSU3 configuration page have five part, it's include Set 、Upgrade 、Other 、Language and Account.

There are 3 account groups and have different access permissions, below table list default account in each group.

Group	Account	Password	Permission
Aamin	zerova	t6sj4j83	Full permission
Manager	manage	ej03xu3m06	Network / Backend configurable
User	monitor	ru0 dj/4	Configuration read only

Form 3-1 Account group and default password table

Each part detail function will descript in below chapter.

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3.1.Set

3.1.1. System

Each partial information display depend on CHARGER model name.

User can read and write CHARGER system information, all item detail as below image and table :

EV Quick Charger

SET UPGRADE OTHER LANGUAGE ACCOUNT

System

System Information

Model Name

AXLU11001D3PG

Serial Number

D19S20001A0

System ID

AXLU11001D3PGD19S20001A0

System Date Time

2025-03-19 08:01:29

☐ Factory Default Configuration

Authentication

Enable

☐ Authorized By EVCCID

Input Voltage R

220.7 volt

Input Voltage S

220 volt

Input Voltage T

220.7 volt

Rfid Card Num Endian

little endian

Rating Current

48 amp

☐ APP ☒ QR Code ☐ RFID

QR Code Made Mode

default

LED Configuration Mode

Sunrise/Sunset Event (Auto)

LED Sunrise Intensity(%)

25

LED Sunset Intensity(%)

10

LCM Configuration Mode

Sunrise/Sunset Event (Auto)

LCM Sunrise Intensity(%)

25

LCM Sunset Intensity(%)

10

Version Information

Set

Image 3-1 System information page

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System Information			
Field Name	Description	Attributes	
		Read	Write
Model Name	CHARGER model name	V	
Serial Number	CHARGER serial number	V	
System ID	Default value is model name and serial number	V	V
System Date Time	CHARGER system date & time	V	V
Ac Phase Loss Policy	The charger will check this setting to see if go into charging state under phase loss - De-rating - Stop charging	V	V
Factory Default Configuration	Request CHARGER restore to factory default value	V	V
Authentiaction	Authorization mode - enable - disable	V	V
Authorized By EVCCID	Auto read EV MAC address as start user idtag, only effect on CCS connector. -enable -disable	V	V
Input VoltageR	AC input vontage R phase	V	
Input VoltageS	AC input vontage S phase	V	
Input VoltageT	AC input vontage T phase	V	
System Fan Speed	System fan speed	V	
Rfid Card Num Endian	RFID number endianness - Little endian - Big endian	V	V
Rating Current	CHARGER rating current	V	
Ac Rating Current	CHARGER AC rating current for third connector	V	
APP	Using APP or not on LCM UI display	V	V
QR Code	Using QR Code or not on LCM UI display	V	V
RFID	Using RFID or not on LCM UI display	V	V

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QR Code Made Mode	Using QR Code - default - customized	V	V
QR Code Content	QR Code Made Mode choosing "customized"	V	V
LED Configuration Mode	LED Intensity mode - Default (100%) - Fixed Maximum (0~100%) - Sunrise/Sunsev Event(Auto) - Sunrise/Sunsev Event(Manual)	V	V
LCM Configuration Mode	LCM Intensity mode - Default (100%) - Fixed Maximum (0~100%) - Sunrise/Sunsev Event(Auto), (need internet access ability) - Sunrise/Sunsev Event(Manual), (UTC+0 timezone)	V	V
Log Include Pcap File	Log Download include CCS pcap file(DC charger only) - Exclude - Include	V	V

Form 3-2 System Information item

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EV Quick Charger

SETUPGRADEOTHERLANGUAGE

System

System Information +

Version Information -

Csu Boot Load Fw Rev01.01

Csu Kernel Fw Rev4.9.59-ga75d8e9305

Csu Root Fs Fw RevD0.01.04.20.03.00

Csu Prim Fw Rev

Psu Prim Fw Rev

Psu Sec Fw Rev

Fan Module Fw Rev

Relay Module Fw Rev

Connector1(CHAdEMO) Fw Rev

Connector2(CCS) Fw Rev

Led Module Fw Rev

Set

Image 3-2 Version information page

Version Information			
Field Name	Description	Attributes	
		Read	Write
Csu Boot Load Fw Rev	CSU bootloader firmware version	V	
Csu Kerne lFw Rev	CSU kernel firmware version	V	
Csu RootFs Fw Rev	CSU RootFileSystem firmware	V	

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	version		
Csu Prim Fw Rev	Main board MCU firmware version	√	
Psu Prim Fw Rev	PSU primary firmware version (only display on DC model)	√	
Psu Sec Fw Rev	PSU secondary firmware version (only display DC model)	√	
Fan Module Fw Rev	FAN module firmware version (only display DC model)	√	
Relay Module Fw Rev	Relay module firmware version (only display DC model)	√	
Telcoom Modem Fw Rev	4G module firmware version (only display support 4G model)	√	
Connector1 Fw Rev	Connector1 module firmware version(only display DC model)	√	
Connector2 Fw Rev	Connector2 module firmware version(only display DC model with dual connectors)	√	
Led Module Fw Rev	4G module firmware version (only display DC model)	√	

Form 3-3 Version Information item

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SETUPGRADEOTHERLANGUAGEACCOUNT

System

System Information+

Version Information+

UI LCD Setting-

Default Language

日本語

2nd

3rd

4th

5th

6th

7th

8th

9th

Set



Image 3-3 UI LCD language setting page

LCD language setting			
Field Name	Description	Attributes	
		Read	Write
Default Language	LCD default display language	V	V
2 nd	Language selection item on UI	V	V
3 rd	Language selection item on UI	V	V

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4 th	Language selection item on UI	V	V
5 th	Language selection item on UI	V	V
6 th	Language selection item on UI	V	V
7 th	Language selection item on UI	V	V
8 th	Language selection item on UI	V	V
9 th	Language selection item on UI	V	V

Form 3-4 LCD language setting item

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3.1.2. Charging



Charging Configuration

Max Charging Energy kWh

0

'0' means unlimit

Max Charging Power kW

100

Max Charging Current amp

0

'0' means unlimit

Max Charging Duration minutes

0

'0' means unlimit

Max Charging Soc percent

0

'0' means unlimit

Stop Charging By Button

Enable

Fan Control Policy

Auto

Local White Card0

Local White Card1

Local White Card2

Local White Card3

Local White Card4

Local White Card5

Local White Card6

Local White Card7

Local White Card8

Local White Card9



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15118

Enable

PnC

Disable

Contract Install/Update

Disable

VDV261

Enable

VDV261 Customized Ipv6 Address

VDV261 Customized Ipv6 Gateway

VDV261 Customized Ipv6 Dns

☐ Billing

Image 3-4 Charging reievant information page

Charging Relevant Parameters			
Field Name	Description	Attributes	
		Read	Write
Max Charging Energy	The maximum charging energy per charge, charger will stop charging if present output energy reach to this threshold 0 : no limit 1~65535 KWH	V	V
Max Charging Power	The maximum charging power for each charge 0 : Rating value 1~rating value kW	V	V
Max Charging Current	The maximum charging current for each DC charge 0 : Rating value 1~rating value Amp	V	V
AC Max Charging Current	The maximum charging current for AC gun on DC charge 0 : Rating value	V	V

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	1~rating value Amp		
Max Charging Duration	The maximum charging duration for each charge 0 : no limit 1~65535 minutes	√	√
Stop Charging By Button	The state of stopping charging button - disable - enable	√	√
LocalWhiteCard0 LocalWhiteCard9	This array stores the ID of administrator RFID card, charger will bypass the user authentication if administrator RFID card be swiped.	√	√
15118	ISO-15118 feature support	√	√
PnC	PnC feature support	√	√
Contract Install/Update	PnC contract install/update feature support	√	√
VDV261	Share network to EV by charging cable	√	√
VDV261 Customized Ipv6 Address	VDV261 customization IPv6 address for EV usage	√	√
VDV261 Customized Ipv6 Gateway	VDV261 customization IPv6 gateway for EV usage	√	√
VDV261 Customized Ipv6 Dns	VDV261 customization IPv6 DNS for EV usage	√	√
Multi Connector Charging Mode	Synchronous: Whole connector charging at same time Sequential: First in First out	√	√
R2R	Charging session restart by CP BCB toggle.	√	√
Billing	Using billing mode or not	√	√
Currency	Choosing currency when using billing mode	√	√
Fee0 Fee23	This array stores the fee with every hours.	√	√

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Form 3-5 Charging Configuration item

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SETUPGRADEOTHERLANGUAGE

Charging

Charging Configuration

Connector1 Charging Information(CHAdemo)

User Id

Start Date Time

Stop Date Time

System Status

booting

Connector Temperature

0 °C

Present Charging Voltage

0 volt

Present Charging Current

0 amp

Present Charging Power

0 kW

Present Charging Energy

0 kWh

Present Charging Duration

0 seconds

Remain Charging Time

0 seconds

EV Battery Max Voltage

0 volt

EV Battery Target Voltage

0 volt

EV Battery Soc

0

Connector2 Charging Information(CCS)

Set

Image 3-5 Charging information page

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Each partial information display depend on CHARGER model name.

Charging InformationX			
Field Name	Description	Attributes	
		Read	Write
User Id	The user use this ID to trigger charging event, it can be RFID card number, OCPP IdTag, etc.	V	
Star tDate Time	Charging cycle start time	V	
Stop Date Time	Charging cycle stop time	V	
System Status	CHARGER system present status	V	
Connector Temperature	Connector temperature Unit : °C	V	
Present Charging Current	Present charging current Unit : A	V	
Present Charging Power	Present charging power Unit : KW	V	
Present Charged Energy	Present charging energy Unit : KWh	V	
Present Charged Duration	Present charging duration Unit : second	V	
Ev Battery Max Voltage	EV battery maximum voltage	V	
Ev Battery Target Voltage	EV battery target voltage	V	
Ev Battery Soc	Present EV battery SOC	V	
Totalize Power Consumption	Totalize Power Consumption Unit : KWh	V	

Form 3-6 Charging Information item

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SETUPGRADEOTHERLANGUAGE

Charging

Charging Configuration+

Connector1 Charging Information(AC)+

V2G Certificate-

V2G Private Key file sha256 hash: (File not found)

Select file...Browse ...

V2G Certificate Chain file sha256 hash: (File not found)

Select file...Browse ...

V2G Root Certificate file sha256 hash: (File not found)

Select file...Browse ...

MO Root Certificate file sha256 hash: (File not found)

Select file...Browse ...

Image 3-6 V2G certificate page

V2G certificate			
Field Name	Description	Attributes	
		Read	Write
V2G Private Key	Private key for encrypt message between EVSE/EV, key type is ECDSA and format is x509.		V
V2G Certificate Chain	Certificate chain for decrypt message between EVSE/EV, certificate chain format is x509.		V
V2G Root Certificate	RootCA for verify certificate chain which come from EV side, CA format is x509.		V
MO Root Certificate	Mobility operator certificate for verify MO-sub certificate which		V

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	from EV.		
--	----------	--	--

Form 3-7 V2G certificate item

3.1.3. Network

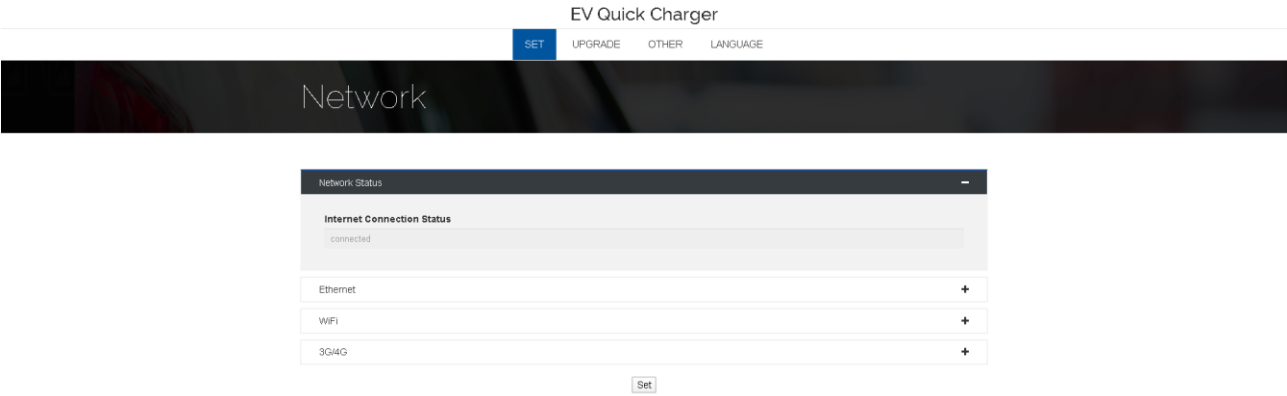


Image 3-7 Network status information page

Network Status					
Field Name	Description			Attributes	
				Read	Write
Internet Connection Status	CHARGER status	internet	connection	V	

Form 3-8 Network status information item

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Network

DashboardLogout

Network Status

Ethernet

DHCP Client

enable

MAC Address

A8:10:87:A9:AB:4E

IP Address

192.168.100.121

Submask Address

255.255.255.0

Gateway Address

192.168.100.1

☒ Wifi to Ethernet Bridge

NAT 4G Interface

ppp0

WiFi

3G/4G

Firewall

Set

Image 3-8 Ethernet-0 interface information page

Ethernet Interface			
Field Name	Description	Attributes	
		Read	Write
Dhcp Client	Ethernet DHCP client ability selection	V	V
Mac Address	Ethernet MAC address	V	
Ip Address	Ethernet IP address	V	V
Submask Address	Ethernet submask address	V	V
Gateway Address	Ethernet gateway address	V	V
Wifi to Ethernet	Bridge ethernet as lan port of WIFI	V	V

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Bridge	AP		
NAT 4G Interface	Select interface to share internet connection ability Ppp0: 1 st 4G module Ppp1: 2 nd 4G module	V	V

Form 3-9 Ethernet information item

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SETUPGRADEOTHERLANGUAGEACCOUNT

Network

Network Status+

Ethernet+

WiFi-

Mode

station

SSID

ASUS_AP

Scan

Password

Client EAP-TLS Identity

asus_ap

Client EAP-TLS private key file sha256 hash: (01e01b979c57a8b9016b216f5abd2b82d10e5a684ef7f4528497673361d67d8d)

Select file...

Browse ...

Client EAP-TLS certificate file sha256 hash: (83d51e3bd6c9e3d18e96097fe698a8f16734cfd5eb1b29b4b34e7569b01557c)

Select file...

Browse ...

Client EAP-TLS root certificate file sha256 hash: (79ae869fe89b9ae51b189e4c3ee012902678f3c026e47c58bfc9c08a9fb5ba2d)

Select file...

Browse ...

Wifi Target Bssid Mac

RSSI

-32 dBm

DHCP Client

enable

MAC Address

D4:CA:6E:AD:99:72

IP Address

192.168.100.184

Submask Address

255.255.255.0

Gateway Address

192.168.100.1

Network Connection Status

Disconnected

3G/4G+

Firewall+

Set

Image 3-9 Wifi module information page

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WiFi Module			
Field Name	Description	Attributes	
		Read	Write
Mode	Wi-Fi module mode - Disable - Station - Access Point	√	√
SSID	1. AP : Fix use modelname & SN 2. Station : Target connect SSID	√	√
Password	The password or preshare key for client to do login	√	√
Client EAP-TLS Identity	WPA enterprise EAP-TLS login identity	√	√
Client EAP-TLS private key	WPA enterprise EAP-TLS login private key		√
Client EAP-TLS certificate	WPA enterprise EAP-TLS login certificate		√
Client EAP-TLS root certificate	WPA enterprise EAP-TLS login root certificate		√
Wifi Target Bssid Mac	Bonding AP MAC address Sample: AA:BB:CC:DD:EE:FF	√	√
RSSI	the received signal strength of WiFi connection	√	
Dhcp Server	DHCP server will be ran under infrastructure server mode if enabled	√	√
DhcpClient	DHCP client will be ran via WiFi module if enabled	√	√
Mac Address	Wi-Fi MAC address	√	
Ip Address	Wi-Fi IP address	√	√
Submask Address	Wi-Fi submask address	√	√
Gateway Address	Wi-Fi gateway address	√	√
Network Connection Status	Wi-Fi network connection status	√	

Form 3-10 WiFi module information item

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SETUP

SETUP

UPGRADE

OTHER

LANGUAGE

Network

Network Status

Ethernet

WiFi

3G/4G

Mode

Disable

APN

Internet

Network Type

Auto

RSSI

0 dBm

Chap Pap Id

Chap Pap Pwd

Modem IMEI

SIM IMSI

SIM ICCID

SIM Status

no SIM card is found

Modem Mode

No services

IP Address

Network Connection Status

Disconnected

Firewall

Set

Image 3-10 3G/4G module information page

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3G/4G Module			
Field Name	Description	Attributes	
		Read	Write
Mode	Telecom module mode - Disable - Enable	V	V
APN	The APN which charger want to connected	V	V
Network Type	-Auto -4G -3G -2G	V	V
Rssi	The signal strength of 3G/4G connection	V	
Chap Pap Id	The login ID for CHAP/PAP authentication	V	V
Chap Pap Pwd	The login password for CHAP/PAP authentication	V	V
Modem IMEI	The IMEI code of this telcom modem	V	
SIM IMSI	The IMSI code of this SIM card	V	
SIM ICCID	Telcom SIM ICCID	V	
SIM Status	Present SIM card status	V	
Modem Mode	Present telcom system mode through this modem	V	
IP Address	Telcom IP address	V	
Network Connection Status	Telcom network connection status	V	

Form 3-11 3G/4G module Information item

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EV Quick Charger

SETUP

SETUP

UPGRADE

OTHER

LANGUAGE

Network

Network Status

+

Ethernet

+

WiFi

+

3G/4G

+

Firewall

-

Mode

Enable

▼

Firewall Accept Addr0

Firewall Accept Addr1

Firewall Accept Addr2

Firewall Accept Addr3

Firewall Accept Addr4

Firewall Accept Addr5

Firewall Accept Addr6

Firewall Accept Addr7

Firewall Accept Addr8

Firewall Accept Addr9

Set

Image 3-11 Firewall information page

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Firewall			
Field Name	Description	Attributes	
		Read	Write
Mode	Firewall module mode - Disable - Enable, deny all packet except charger build-in use server address and address on white list.	V	V
Firewall Accept Addr0	Server address whitelist 0	V	V
Firewall Accept Addr1	Server address whitelist 1	V	V
Firewall Accept Addr2	Server address whitelist 2	V	V
Firewall Accept Addr3	Server address whitelist 3	V	V
Firewall Accept Addr4	Server address whitelist 4	V	V
Firewall Accept Addr5	Server address whitelist 5	V	V
Firewall Accept Addr6	Server address whitelist 6	V	V
Firewall Accept Addr7	Server address whitelist 7	V	V
Firewall Accept Addr8	Server address whitelist 8	V	V
Firewall Accept Addr9	Server address whitelist 9	V	V

Form 3-12 Firewall Information item

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3.1.4. Backend

EV Quick Charger

SETUPGRADEOTHERLANGUAGEACCOUNT

Backend

Common Information

Backend Connection Timeout

300

seconds

Support IPv6

Disable

Offline Policy

local list

Offline Max Charge Energy kWh

0

Offline Max Charge Duration minutes

0

Offline Max Charge Current amp

0

Modbus TCP Server

+

OCPP Backend

+

IEEE 2030.5

+

Set



Image 3-12 Backend common information page

Common Information			
Field Name	Description	Attributes	
		Read	Write
Backend Connection Timeout	Timeout limit of connecting to backend	√	√

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Suport IPv6	OCPP ipv6 client feature - Disable - Enable	V	V
Offline Policy	CHARGER offline charging policy - Local list - Free charging - No charging (End by user) - No charging (End immediately)	V	V
Offline Max Charge Energy	CHARGER offline charging energy 0 : MaxChargingEnergy 1~65535 KWH	V	V
Offline Max Charge Duration	CHARGER offline charging duration 0 : MaxChargeDuration 1~65535 Minutes	V	V
Offline Max Charge Power	CHARGER offline mximum output power , only DC model 0: MaxChargePower 1~65535 kWh	V	V
Offline Max Charge Current	CHARGER offline mximum output current , only AC model 0: MaxChargePower 1~65535 amp	V	V

Form 3-13 Common information item

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EV Quick Charger

SETUPGRADEOTHERLANGUAGEACCOUNT

Backend

Common Information+

Modbus TCP Server+

OCPP Backend-

OCPP Connection Status

Connected

Central System URL

wss://charge.zerovatech.com/server/

Support V2.0.1

Disable

Charge Box Id

AXLU111001D1P0D19520001A0

Charge Point Vendor

Philhong Technology

OCPP Security Profile

None security

Manufacturer Root Certificate file sha256 hash: (a4d6770352e1cce9920e1307a329d81b3360a17ed739f4a7e93cd3e9990e31fa)

Select file...Browse ...

Receipt URL

Maintain Server Connection Status

Connected

Maintain Server URL

wss://foluswen.com:8443/steve/websocket/CentralSystemService/

Maintain Server Security Profile

None security

Local Loading Balance+

IEEE 2030.5+

Set

Image 3-13 Ocpp backend information page

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OCPP Backend			
Field Name	Description	Attributes	
		Read	Write
Ocpp Connection Status	The connection status of OCPP backend	V	
Central System URL	The OCPP server url which include port which charger want to connected. ws: non-secure OCPP 1.6-J wss: secure OCPP 1.6-J Ex: ws://ocpp.phihong.com.tw:80/ wss://ocpp.phihong.com.tw:443/	V	V
Charge BoxId	The charge box ID will use to connect to OCPP server	V	V
Charge Point Vendor	Charger vender name	V	V
Ocpp Security Profile	Ocpp Security Profile - None security - Unsecured Transport with Basic Atuentication - TLS with Basic Authentication - TLS with Client Side Certificates	V	V
Ocpp Security Password	Displayed when "Ocpp Security Profile" is 「Unsecured Transport with Basic Atuentication」 or 「TLS with Basic Authentication」	V	V
Receipt URL	Receipt QR code URL content, if filled out value when charging session end LCM will display QR Code for end user scan to get receipt info.	V	V
Maintain Server Connection Status	The connection status of maintain backend	V	
Maintain Server URL	The maintain server url which include port which charger want	V	V

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	to connected. ws: non-secure OCPP 1.6-J wss: secure OCPP 1.6-J Ex: ws://ocpp.phihong.com.tw:80/ wss://ocpp.phihong.com.tw:443/		
Maintain Server Security Profile	Ocpp Security Profile - None security - Unsecured Transport with Basic Authentication - TLS with Basic Authentication	V	V

Form 3-14 OCPP backend information item

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EV Quick Charger

SETUPGRADEOTHERLANGUAGEACCOUNT

Backend

Common Information+

Modbus TCP Server+

OCPP Backend+

Local Loading Balance-

Mode

Server

Power Sharing Capacity^W

0

^W means follow rating power from model name

IEEE 2030.5+

Set

Image 3-14 Local loading balance information page

Local loading balance			
Field Name	Description	Attributes	
		Read	Write
Local Loading Balance	Local Loading Balance - Disable - Client Server	V	V
Power Sharing Capacity	Power capacity when local loading balance configure as server.	V	V
Power Sharing Server IP	Target server IP when local loading balance configure as client	V	V

Form 3-15 Local loading balance information item

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EV Quick Charger

SETUPGRADEOTHERLANGUAGE

Backend

Common Information+

OCPP Backend+

TTIA-

TTIA

Yes

Server Address

http://127.0.0.1

Server Port

65535

Bus Vender Id

65535

Equipment Provider

飛宏科技

TransportationCompanyNo

255

Charge Box Id

255

EVSE Station

飛宏科技站

Set

Image 3-15 TTIA backend information page

TTIA Backend(for CNS only)			
Field Name	Description	Attributes	
		Read	Write
TTIA	TTIA setting enable or disable	V	V
Server Address	The TTIA server url which charger want to connected.	V	V
Server Port	The TTIA server port which charger want to connected.	V	V
Bus Vender Id	Bus vender ID for TTIA	V	V
Equipment Provider	EVSE provide vender	V	V
Transportation Company No	TTIA info transportation company id	V	V
Charge Box Id	EVSE ID for TTIA	V	V
EVSE Station	EVSE locate station	V	V

Form 3-16 TTIA backend information item

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EV Quick Charger

SETUPGRADEOTHERLANGUAGEACCOUNT

Backend

Common Information+

Modbus TCP Server-

Mode

Enable

Password Verify

Yes

Password

testpass

TLS encryption

No

OCPP Backend+

IEEE 2030.5+

Set

Image 3-16 Modbus TCP Server page

MODBUS Tcp Server			
Field Name	Description	Attributes	
		Read	Write
Mode	Disable : Modbus TCP server shutdown. Enable : Modbus TCP server will be host on port 502 and only allow one connection in.	V	V
Password Verify	Disable : Access any modbus register directly. Enable : Before access modbus register must write password hash to password register for login	V	V

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	verification, if verification result pass, charger will fill value to all modbus register		
Password	1. This effect if password verify as enable. 2. Password must hash by sha256 in lower case.	V	V
TLS encryption	Disable : Modbus TCP in transparentcy mode Enable : Modbus TCP in TLS encrypted mode	V	V

Form 3-17 Modbus TCP Server item

Modbus register information please reference appendix.

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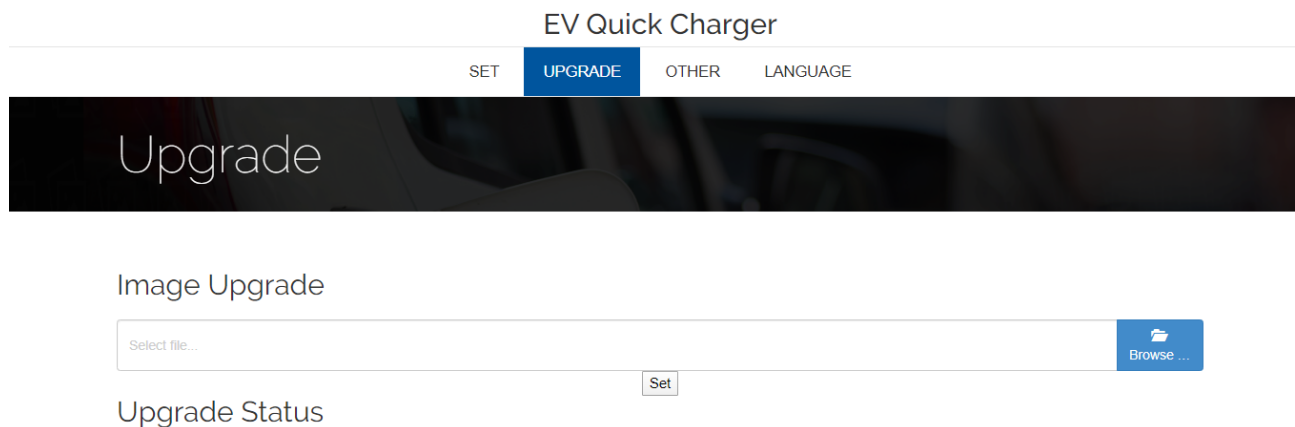
(2025/09/30)

3.2. Upgrade

All of firmware image can be upload from this page to upgrade, charger will auto check image header & CRC32 before upgrade.

If image valid pass, charger will upgrade in IDLE mode and reboot after upgraded.

Charger indicator in upgrading please reference each model system specification.



The screenshot shows the 'EV Quick Charger' interface. At the top, there is a navigation bar with four tabs: 'SET', 'UPGRADE' (which is highlighted in blue), 'OTHER', and 'LANGUAGE'. Below the navigation bar is a large dark banner with the word 'Upgrade' in white. Underneath the banner, the text 'Image Upgrade' is displayed. Below this text is a file selection area consisting of a text input field with the placeholder 'Select file...', a 'Browse ...' button with a folder icon, and a 'Set' button. At the bottom of the form, the text 'Upgrade Status' is visible.

Image 3-17 Charger upgrade set page

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3.3. Other

3.3.1. Download Log

After click download tab item, charger will pack all log file into a compress zip file and show dialog which save destination select.

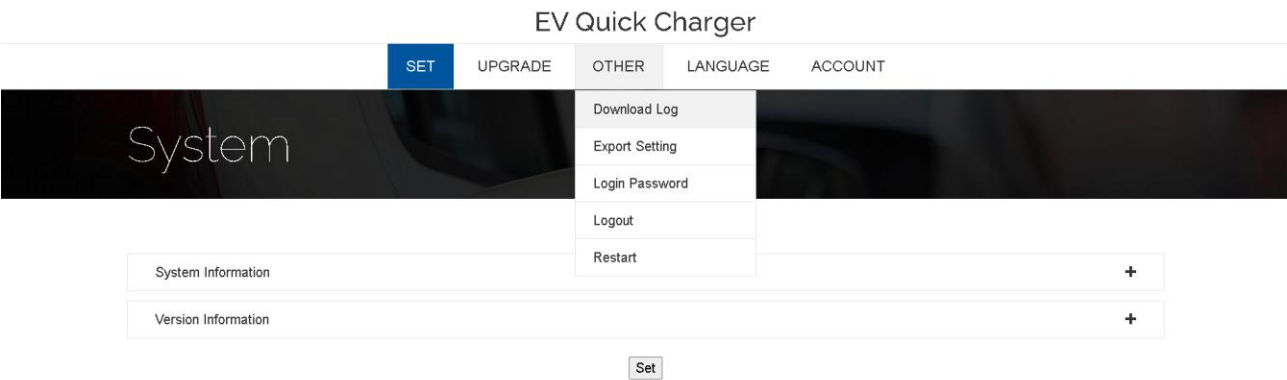


Image 3-18 Download page

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3.3.2. Login password

After change configuration web page login password, user need to use new password to login on next time.

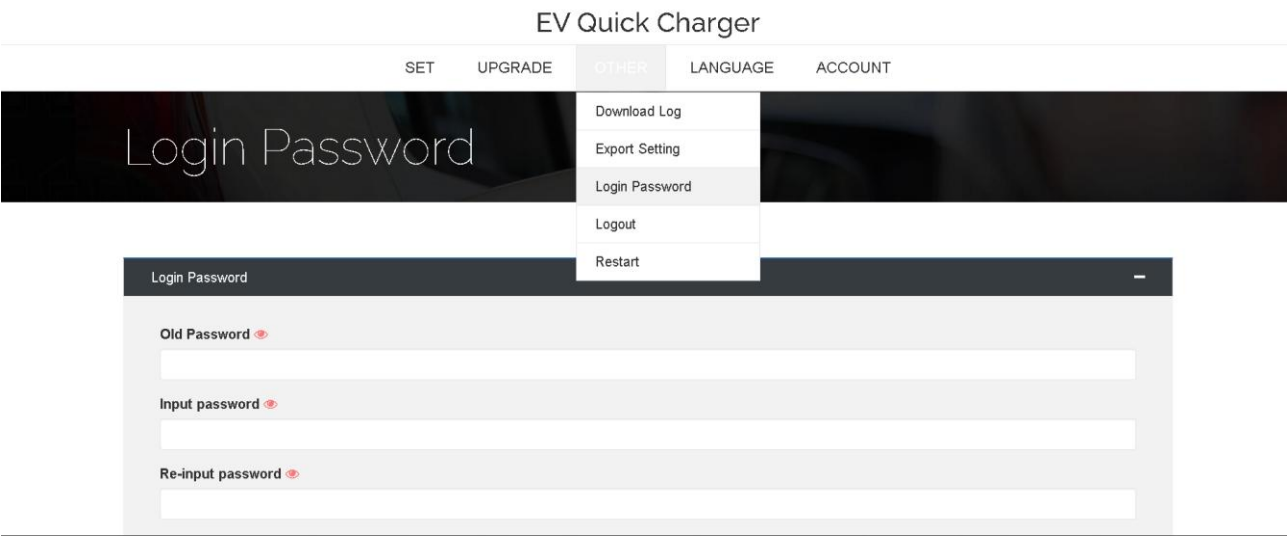


Image 3-19 Configuration login password set page

ZeroVa Tech

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3.3.3. Export Setting

After click Export Setting item, charger will export all configuration value to a bin file and show dialog which save destination select.

Export setting file only can be import to same model name charger.

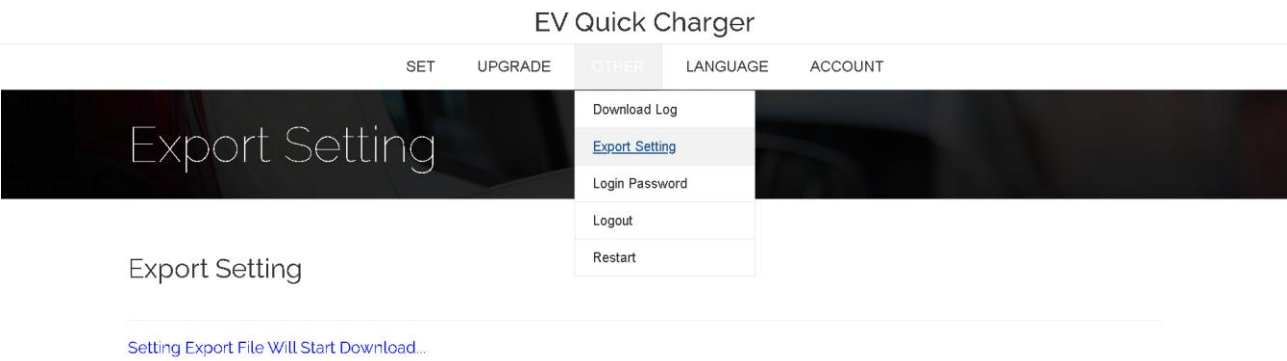


Image 3-20 Export setting page

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3.3.4. Restart

After click restart item charger will end charging session and reboot controller.

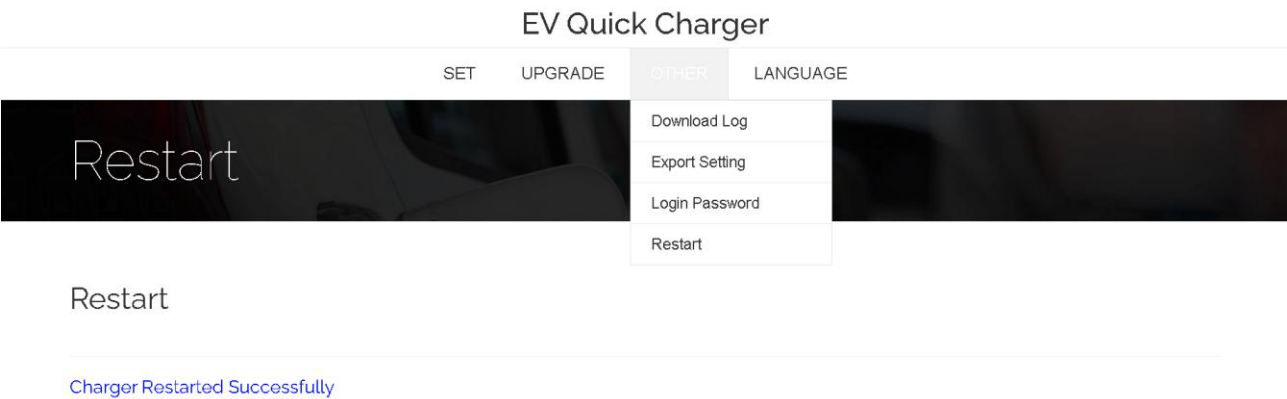


Image 3-21 Restart page

3.4. Language

Configuration web page have three language interface can be set.



Image 3-22 Configuration UI language set page

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3.5.Account

This page will appear when user account role is admin.

There are three role configurations:

- Admin : Full configuration function.
- Manager : Network / Backend configuration 、 firmware upgrade and log file download function.
- User : Read only for all configuration item and download log file.

Password must match complexity principle ◦

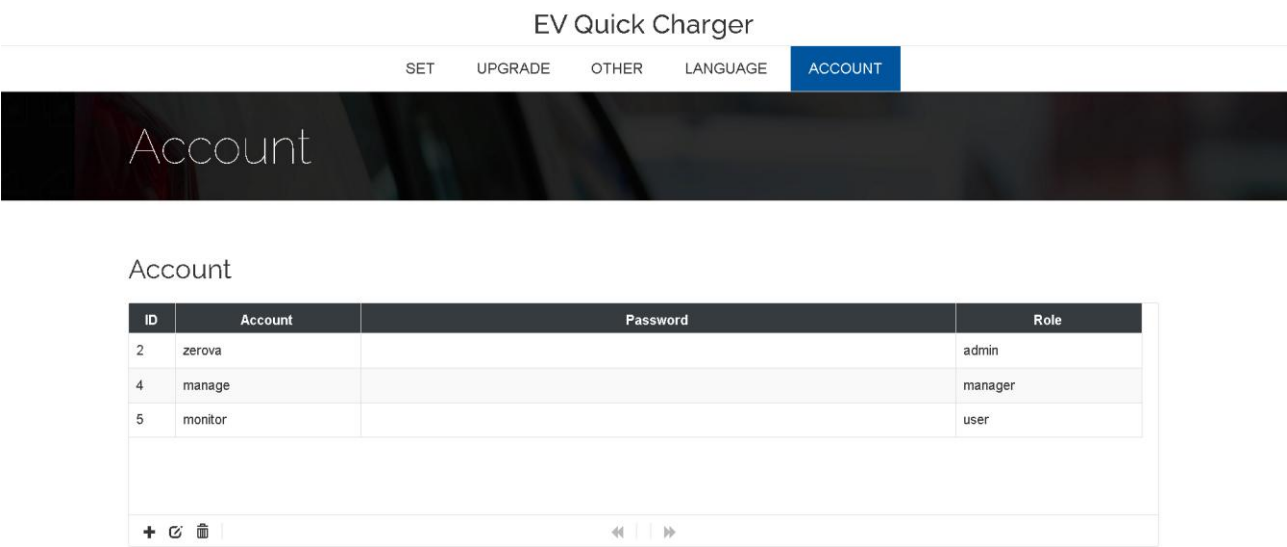


Image 3-23 Account set page

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4. Appendix

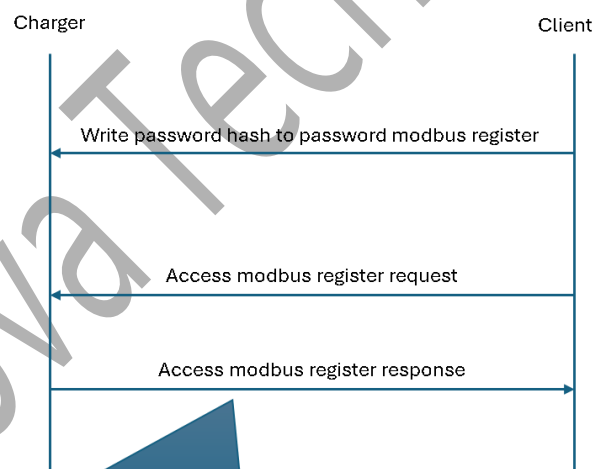
4.1. Modbus information

Data endian default is little endian unless specified on each register description

4.1.1. Modbus server parameters

Parameters of CSUs	
Modbus port	502
Size of nb bits	39000
Size of nb input bits	39000
Size of nb registers	39000
Size of nb input registers	39000

Form 4-1 Modbus TCP Server parameters



1. Password verify configured as disable, charger will response real value to register.
2. Password verify configured as enable, client must write correct password hash to password register then access other register normally.

Image 4-1 Modbus access description

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4.1.2. Coil table

Following is the summary table of coil, which is using coil (function code is 0x01/0x05). Please refer start from chapter 4.1.2.1 for the detail.

Bit group	Address of coil	Name	Type
Common operation bits	0	Write EVSE configuration value request	W
	1	EVSE reboot request	W
	2	String data MSB request	W
	3	String data MSW request	W
Connector-id operation bits (id start since 1)	(id*1000) + 0	Coonector-1 start charging request	W
	(id*1000) + 1	Coonector-1 stop charging request	W
	(id*1000) + 2	Coonector-1 operative request	W
	(id*1000) + 3	Coonector-1 inoperative request	W
	(id*1000) + 4	Coonector-1 plugged count clear request	W
	...	Reserve	
	(id*1000) + 999		

Form 4-2 Modbus coil table

4.1.2.1. Write EVSE configuration value request

Coil usage	
Description	External control system request EVSE write currently configuration value on holding register to NAND flash.
Coil address	0
Permission	Write only

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4.1.2.2. EVSE reboot request

Coil usage	
Description	External control system request EVSE perform system reboot.
Coil address	1
Permission	Write only

4.1.2.3. String data byte swap request

Coil usage	
Description	External control system request EVSE put string data with byte swap.
Coil address	2
Permission	Write only

4.1.2.4. String data word swap request

Coil usage	
Description	External control system request EVSE put string data with word swap.
Coil address	3
Permission	Write only

4.1.2.5. Connector start charging request

Coil usage	
Description	External control system request connector start charging session.
Coil address	(id*1000)
Permission	Write only

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4.1.2.6. Connector stop charging request

Coil usage	
Description	External control system request connector stop charging session.
Coil address	(id*1000)+1
Permission	Write only

4.1.2.7. Connector operative request

Coil usage	
Description	External control system request connector operative.
Coil address	(id*1000)+2
Permission	Write only

4.1.2.8. Connector inoperative request

Coil usage	
Description	External control system request connector inoperative.
Coil address	(id*1000)+3
Permission	Write only

4.1.2.9. Connector plugged count clear request

Coil usage	
Description	External control system request to clear connector plugged count.
Coil address	(id*1000)+4
Permission	Write only

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4.1.3. Register table

Following is the summary table of register, which is using holding register (function code is 0x03/0x06). Please refer start chapter 4.1.3.1 for the detail.

Data endian default is little endian unless specified on each register description

Register group	Address of holding Register	Name	Type
EVSE info	0	Model name	R
	32	Serial number	R
	64	Login password	W
	96	Present AC input voltage L1	R
	98	Present AC input voltage L2	R
	100	Present AC input voltage L3	R
	102	Present DC input voltage	R
EVSE configuration	564	Authentication enable	R/W
	565	Authentication by EVCCID	R/W
	566	Maximum charging energy (kWh)	R/W
	567	Maximum charging power (kW)	R/W
	568	Maximum charging current (A)	R/W
	569	Maximum charging duration (Minutes)	R/W
	570	RFID endian	R/W
	571	ISO-15118 enable	R/W
	572	ISO-15118 PnC enable	R/W
Connector-id info (id start from 1)	(id*1000) + 0	Connector system status	R
	(id*1000) + 1	Connector plug status	R
	(id*1000) + 2	Present charging session SOC (%)	R
	(id*1000) + 3	Present charging output power (kW)	R
	(id*1000) + 5	Present charging output voltage DC (V)	R
	(id*1000) + 7	Present output voltage AC L1 (V)	R
	(id*1000) + 9	Present output voltage AC L2 (V)	R
	(id*1000) + 11	Present output voltage AC L3 (V)	R
	(id*1000) + 13	Present output current DC (A)	R
	(id*1000) + 15	Present output current AC L1 (A)	R
	(id*1000) + 17	Present output current AC L2 (A)	R
	(id*1000) + 19	Present output current AC L3 (A)	R

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	(id*1000) + 21	Present charged energy (kWh)	R
	(id*1000) + 25	Present charged duration (Second)	R
	(id*1000) + 27	Present charging session remain time (Second)	R
	(id*1000) + 29	Present charging session idtag	R
	(id*1000) + 39	Connector plugged count	R
	(id*1000) + 41	Connector Accumulate output energy (KWH)	R
	(id*1000) + 50	Connector status code in ASCII format	R
	(id*1000) + 90	Connector status code in integer format	R
	(id*1000) + 100	Connector temperature	R
	(id*1000) + 101	Connector charging session EVCCID	R
	(id*1000) + 111	Connector ID	R
Key parts information (id / sub_id start from 1)	20000	Charger(cabinet) AC breaker usage time	R
	20002	Charger(cabinet) AC contactor cycle count	R
	20004	Charger(cabinet) filter usage time	R
	21000	Charger(cabinet) output relay quantity (id)	R
	(id*100) + 21000	K1 output relay usage count	R
	(id*100) + 21002	K2 output relay usage count	R
	24000	Charger(cabinet) bridge relay quantity (id)	R
	(id*100) + 24000	P bridge relay usage count	R
	(id*100) + 24002	N bridge relay usage count	R
	25000	Dispenser quantity (id)	R
	(id*1000) + 25000	Dispenser AC breaker usage time	R
	(id*1000) + 25002	Dispenser AC contactor cycle count	R
	(id*1000) +	Dispenser filter usage time	R

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	25004		
	(id*1000) + 25500	Dispenser output relay quantity (sub_id)	R
	(id*1000) + (sub_id*100) + 25500	Dispenser K1 output relay usage count	R
	(id*1000) + (sub_id*100) + 25502	Dispenser K2 output relay usage count	R

Form 4-3 Modbus register table

Hierarchy		Component Name	Register address
Charger (Cabine)		AC breaker	20000
		AC contactor	20002
		filter	20004
		Output relay set quantity	21000
	Output relay (id start from 1 to output relay set quantity)	Connector-1 K1 relay	(id*100)+21000
		Connector-1 K2 relay	(id*100)+21002
		...	(id*100)+21000
		...	(id*100)+21002
		Bridge relay set quantity	24000
	Bridge relay (id start from 1 to Bridge relay quantity)	Bridge-1 P relay	(id*100)+24000
		Bridge-1 N relay	(id*100)+24002
		...	(id*100)+24000
		...	(id*100)+24002
		Dispenser quantity	25000
	Dispenser (id start from 1 to dispenser quantity)	AC breaker	(id*1000)+25000
		AC contactor	(id*1000)+25002
		filter	(id*1000)+25004
		Output relay set quantity	(id*1000)+25500
		Output relay (sub_id start from 1 to output relay set quantity)	Connector-1 K1 relay (id*1000)+(sub_id*100)+25500
		Connector-1 K2 relay	(id*1000)+(sub_id*100)+25502
		...	(id*1000)+(sub_id*100)+25500
		...	(id*1000)+(sub_id*100)+25502

Form 4-4 Key parts modbus register architecture table

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4.1.3.1. Model name

Register usage	
Description	EVSE model name information
Register address	0
Length	32 word in string
Permission	Read only

4.1.3.2. Serial number

Register usage	
Description	EVSE serial number information
Register address	32
Length	32 word in string
Permission	Read only

4.1.3.3. Login password

Register usage	
Description	Login password before access modbus info must write correct password.
Register address	64
Length	32 word in string
Permission	W
Parameter	This content is password sha256 hash result in lower case

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4.1.3.4. Present input Voltage L1

Register usage	
Description	EVSE AC Input voltage R (kW).
Register address	96
Length	2 word in float
Permission	Read only

4.1.3.5. Present input Voltage L2

Register usage	
Description	EVSE AC input voltage S (kW).
Register address	98
Length	2 word in float
Permission	Read only

4.1.3.6. Present input Voltage L3

Register usage	
Description	EVSE AC input voltage T (kW).
Register address	100
Length	2 word in float
Permission	Read only

4.1.3.7. Present input Voltage DC

Register usage	
Description	EVSE DC input voltage (kW).
Register address	102
Length	2 word in float
Permission	Read only

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4.1.3.8. Authentication enable

Register usage		
Description	Configure EVSE perform to authorize Idtag before start charging session.	
Register address	564	
Length	1 word in integer	
Permission	R/W	
Parameter	0	Disable
	1	Enable

4.1.3.9. Authentication by EVCCID

Register usage		
Description	Configure EVSE perform to get EVCCID from EV as Idtag.	
Register address	565	
Length	1 word in integer	
Permission	R/W	
Parameter	0	Disable
	1	Enable

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4.1.3.10. Maximum charging energy

Register usage		
Description	Configure EVSE each charging session maximum charged energy (kWh).	
Register address	566	
Length	1 word in integer	
Permission	R/W	
Parameter	0	Unlimit
	1~65535	kWh

4.1.3.11. Maximum charging power

Register usage		
Description	Configure EVSE each charging session maximum charging power (kW), only effect on DC model.	
Register address	567	
Length	1 word in integer	
Permission	R/W	
Parameter	Range	Notes
	0	Unlimit
	1~Rating power	kW

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4.1.3.12. Maximum charging current

Register usage		
Description	Configure EVSE each charging session maximum charging current (A).	
Register address	568	
Length	1 word in integer	
Permission	R/W	
Parameter	Range	Notes
	0	Unlimit
	1~Rating current	A

4.1.3.13. Maximum charging duration

Register usage		
Description	Configure EVSE each charging session maximum charged duration (Minute).	
Register address	569	
Length	1 word in integer	
Permission	R/W	
Parameter	Range	Notes
	0	Unlimit
	1~65535	Minutes

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4.1.3.14. RFID endian

Register usage		
Description	Configure EVSE read RFID SN direction	
Register address	570	
Length	1 word in integer	
Permission	R/W	
Parameter	Range	Notes
	0	Little endian
	1	Big endian

4.1.3.15. ISO-15118 enable

Register usage		
Description	Configure EVSE support ISO-15118 communication protocol	
Register address	571	
Length	1 word in integer	
Permission	R/W	
Parameter	Range	Notes
	0	Disable
	1	Enable

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4.1.3.16. ISO-15118 PnC enable

Register usage		
Description	Configure EVSE support ISO-15118 PnC mechanism	
Register address	572	
Length	1 word in integer	
Permission	R/W	
Parameter	Range	Range
	0	Disable
	1	Enable

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4.1.3.17. Connector system status

Register usage		
Description	Connector currently system state	
Register address	(id*1000) + 0	
Length	1 word in integer	
Permission	Read only	
Parameter	Value	Notes
	0	Bootting
	1	Idle
	2	Authorizing
	5	Preparing
	8	Charging
	9	Terminating
	10	Complete
	11	Alarm
	12	Fault
	13	Reservation
	15	Maintain
	19	Upgrade

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4.1.3.18. Connector plug status

Register usage		
Description	Connector and EV socket connection status	
Register address	$(id*1000) + 1$	
Length	1 word in integer	
Permission	Read only	
Parameter	Value	Value
	0	Disconnected
	1	Connected

4.1.3.19. Present charging session SOC

Register usage	
Description	Connector present charging session SOC (%) on EV side
Register address	$(id*1000) + 2$
Length	1 word in integer
Permission	Read only

4.1.3.20. Present charging output power

Register usage	
Description	Connector present charging session output power (kW)
Register address	$(id*1000) + 3$
Length	2 word in float
Permission	Read only

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4.1.3.21. Present charging output voltage DC

Register usage	
Description	Connector present charging session output voltage (V), only effect DC connector.
Register address	(id*1000) + 5
Length	2 word in float
Permission	Read only

4.1.3.22. Present charging output voltage AC L1

Register usage	
Description	Connector present charging session output L1 voltage (V), only effect AC connector.
Register address	(id*1000) + 7
Length	2 word in float
Permission	Read only

4.1.3.23. Present charging output voltage AC L2

Register usage	
Description	Connector present charging session output L2 voltage (V), only effect AC connector with 3 phase power.
Register address	(id*1000) + 9
Length	2 word in float
Permission	Read only

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4.1.3.24. Present charging output voltage AC L3

Register usage	
Description	Connector present charging session output L3 voltage (V), only effect AC connector with 3 phase power.
Register address	(id*1000) + 11
Length	2 word in float
Permission	Read only

4.1.3.25. Present charging output current DC

Register usage	
Description	Connector present charging session output current (A), only effect DC connector.
Register address	(id*1000) + 13
Length	2 word in float
Permission	Read only

4.1.3.26. Present charging output current AC L1

Register usage	
Description	Connector present charging session output L1 current (A), only effect AC connector.
Register address	(id*1000) + 15
Length	2 word in float
Permission	Read only

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4.1.3.27. Present charging output current AC L2

Register usage	
Description	Connector present charging session output L2 current (A), only effect AC connector with 3 phase power.
Register address	(id*1000) + 17
Length	2 word in float
Permission	Read only

4.1.3.28. Present charging output current AC L3

Register usage	
Description	Connector present charging session output L3 current (A), only effect AC connector with 3 phase power.
Register address	(id*1000) + 19
Length	2 word in float
Permission	Read only

4.1.3.29. Present charged energy

Register usage	
Description	Connector present charging session output energy (kWh)
Register address	(id*1000) + 21
Length	2 word in float
Permission	Read only

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4.1.3.30. Present charged duration

Register usage	
Description	Connector present charging session duration (second)
Register address	(id*1000) + 25
Length	2 word in integer
Permission	Read only

4.1.3.31. Present charging session remain time

Register usage	
Description	Connector present charging session remain time (second), only effect DC connector
Register address	(id*1000) + 27
Length	2 word in integer
Permission	Read only

4.1.3.32. Present charging session idtag

Register usage	
Description	Connector present charging session start user, only effect DC connector
Register address	(id*1000) + 29
Length	10 word in string
Permission	Read only

4.1.3.33. Connector plugged count

Register usage	
Description	Connector present plugged accumulative count
Register address	(id*1000) + 39
Length	2 words in integer
Permission	Read only

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4.1.3.34. Connector accumulative output energy

Register usage	
Description	Connector accumulative output energy (kWh)
Register address	(id*1000) + 41
Length	2 words in float
Permission	Read only

4.1.3.35. Connector status code (ASCII)

Register usage	
Description	Connector present status code in ASCII format
Register address	(id*1000) + 50
Length	40 word in string
Permission	Read only
Parameter	If there are two above status code occur mean time, code will distinguish in comma flag.

4.1.3.36. Connector status code (Integer)

Register usage	
Description	Connector present status code in integer format
Register address	(id*1000) + 90
Length	10 word in integer
Permission	Read only
Parameter	If there are two above status code occur mean time, status code will put on next register.

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4.1.3.37. Connector temperature

Register usage	
Description	Connector present temperature in Celsius
Register address	(id*1000) + 100
Length	1 word in integer
Permission	Read only
Parameter	This register only effect with connector support temperature detect feature

4.1.3.38. Connector charging session EVCCID

Register usage	
Description	Connector present charging session EVCCID
Register address	(id*1000) + 101
Length	10 words in string
Permission	Read only
Parameter	This register only effect with CCS connector

4.1.3.39. Connector ID

Register usage	
Description	Connector ID
Register address	(id*1000) + 111
Length	1 word in integer
Permission	Read only

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4.1.3.40. Charger AC breaker use time

Register usage	
Description	Charger (DO/DB/DK/DZ model is cabinet) AC breaker usage time in second, this only for DC model
Register address	20000
Length	2 words in integer
Permission	Read only

4.1.3.41. Charger AC contactor cycle count

Register usage	
Description	Charger (DO/DB/DK/DZ model is cabinet) AC contactor ON/OFF cycle count, this only for DC model
Register address	20002
Length	2 words in integer
Permission	Read only

4.1.3.42. Charger AC filter usage time

Register usage	
Description	Charger (DO/DB/DK/DZ model is cabinet) filter usage time in second, this only for DC model
Register address	20004
Length	2 words in integer
Permission	Read only

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4.1.3.43. Charger output relay set quantity

Register usage	
Description	Charger (DO/DB/DK/DZ model is cabinet) output relay quantity, one output connector will equip K1/K2 relay set, this only for DC model
Register address	21000
Length	2 words in integer
Permission	Read only

4.1.3.44. Charger K1 output relay usage count

Register usage	
Description	Charger (DO/DB/DK/DZ model is cabinet) output relay K1 usage count, this only for DC model, id start from 1 to charger output relay set quantity.
Register address	$(id*100) + 21000$
Length	2 words in integer
Permission	Read only

4.1.3.45. Charger K2 output relay usage count

Register usage	
Description	Charger (DO/DB/DK/DZ model is cabinet) output relay K2 usage count, this only for DC model, id start from 1 to charger output relay set quantity.
Register address	$(id*100) + 21002$
Length	2 words in integer
Permission	Read only

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4.1.3.46. Charger bridge relay quantity

Register usage	
Description	Charger (DO/DB/DK/DZ model is cabinet) bridge relay quantity, one bridge set will equip P/N relay set, this only for DC model
Register address	24000
Length	2 words in integer
Permission	Read only

4.1.3.47. Charger P bridge relay usage count

Register usage	
Description	Charger (DO/DB/DK/DZ model is cabinet) P bridge relay usage count, this only for DC model, id start from 1 to charger bridge relay quantity.
Register address	$(id*100) + 24000$
Length	2 words in integer
Permission	Read only

4.1.3.48. Charger N bridge relay usage count

Register usage	
Description	Charger (DO/DB/DK/DZ model is cabinet) N bridge relay usage count, this only for DC model, id start from 1 to charger bridge relay quantity.
Register address	$(id*100) + 24002$
Length	2 words in integer
Permission	Read only

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4.1.3.49. Dispenser quantity

Register usage	
Description	Dispenser quantity on cabinet, this only for DO/DB/DK/DZ model
Register address	25000
Length	2 words in integer
Permission	Read only

4.1.3.50. Dispenser AC breaker use time

Register usage	
Description	Dispenser AC breaker usage time in second, this only for DO/DB/DK/DZ model, id start from 1 to dispenser quantity.
Register address	$(id * 1000) + 25000$
Length	2 words in integer
Permission	Read only

4.1.3.51. Dispenser AC contactor cycle count

Register usage	
Description	Dispenser AC contactor ON/OFF cycle count, this only for DO/DB/DK/DZ model, id start from 1 to dispenser quantity.
Register address	$(id * 1000) + 25002$
Length	2 words in integer
Permission	Read only

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4.1.3.52. Dispenser AC filter usage time

Register usage	
Description	Dispenser filter usage time in second, this only for DO/DB/DK/DZ model, id start from 1 to dispenser quantity.
Register address	$(id * 1000) + 25004$
Length	2 words in integer
Permission	Read only

4.1.3.53. Dispenser output relay quantity

Register usage	
Description	Dispenser output relay quantity, one output connector will equip K1/K2 relay set, this only for DO/DB/DK/DZ model, id start from 1 to dispenser quantity.
Register address	$(id * 1000) + 25500$
Length	2 words in integer
Permission	Read only

4.1.3.54. Dispenser K1 output relay usage count

Register usage	
Description	Dispenser output relay K1 usage count, this only for DO/DB/DK/DZ model, id start from 1 to dispenser quantity and sub_id start from 1 to dispenser output relay quantity.
Register address	$(id * 100) + 25500$
Length	2 words in integer
Permission	Read only

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4.1.3.55. Dispenser K2 output relay usage count

Register usage	
Description	Dispenser output relay K2 usage count, this only for DO/DB/DK/DZ model, id start from 1 to dispenser quantity and sub_id start from 1 to dispenser output relay quantity.
Register address	$(id * 100) + 25502$
Length	2 words in integer
Permission	Read only